

An $F\ell^1$ -basis need not give η -factorization in $\text{LUC}(\mathcal{A})$

Candidate counterexample for arXiv:2412.15029

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Status. Candidate full counterexample, likely valid, pending expert review.

Source question

Filali and Galindo prove that an $F\ell^1(\eta)$ -basis gives η -factorization in $\mathcal{A}^*$ and one-factorization in $\text{LUC}(\mathcal{A})$. They ask in Remark 4.15(iii) whether it also gives η -factorization in $\text{LUC}(\mathcal{A})$ [1, p. 20].

$$\langle \Psi_\lambda \cdot T, a \rangle = \lim_\gamma \langle \mathbf{a}_{h_\lambda(\gamma)} \cdot T, a \rangle = \lim_\gamma \langle T, a \mathbf{a}_{h_\lambda(\gamma)} \rangle = \langle S_\lambda, a \rangle,$$

as wanted.

- (iii) When a Banach algebra $\mathcal{A}$ is equipped with an $F\ell^1(\eta)$ -base, $\mathcal{A}^*$ has the η -factorization property as it has just been proved, but we have been unable to deduce that $\mathcal{LUC}(\mathcal{A})$ satisfies this property as well, that is for instance when $m = 1$, whether any bounded family of functionals $\{S_\lambda : \lambda < \eta\}$ contained in $\mathcal{LUC}(\mathcal{A})$ can be factorized as $\Psi_\lambda \cdot T$ for some bounded family $\Psi_\lambda \in \mathcal{LUC}(\mathcal{A})^*$ and $T \in \mathcal{LUC}(\mathcal{A})$ for every $\lambda < \eta$.

We can nevertheless show below that the 1-factorization is true in $\mathcal{LUC}(\mathcal{A})$ whenever an $F\ell^1(\eta)$ -base is available in the algebra $\mathcal{A}$, which is enough as we shall see for our study of the algebra and the topological centre of $\mathcal{LUC}(\mathcal{A})^*$. To be able to deduce the η -factorization for $\mathcal{LUC}(\mathcal{A})$ one needs to make sure that the extension T of the operator S defined in (20) is in $\mathcal{LUC}(\mathcal{A})$.

Transcription. In the case $m = 1$, does every bounded family $\{S_\lambda : \lambda < \eta\} \subset \text{LUC}(\mathcal{A})$ admit

$$S_\lambda = \Psi_\lambda \cdot T \quad (\lambda < \eta),$$

where (Ψ_λ) is bounded in $\text{LUC}(\mathcal{A})^*$ and one common T belongs to $\text{LUC}(\mathcal{A})$?

Counterexample

Take the convolution algebra $\mathcal{A} = L^1(\mathbb{R})$. Its compact covering is $\aleph_0$, and source Theorem 6.1 states that $L^1(\mathbb{R})$ has an $F\ell^1(\aleph_0)$ -basis of type 2 [1, p. 28]. Under the standard identification $\text{LUC}(\mathcal{A}) = \text{LUC}(\mathbb{R})$, put

$$S_n(x) = e^{inx}, \quad n = 1, 2, \dots$$

This is a bounded family in $\text{LUC}(\mathcal{A})$, with $\|S_n\|_\infty = 1$.

Proof Intuition

A common factor $T \in \text{LUC}(\mathcal{A})$ imposes a uniform continuity condition on all its bounded left factors: a bounded approximate identity must converge on the whole family uniformly. High-frequency characters violate precisely this condition. Each approximate-identity element has a Fourier transform tending to zero at high frequency, so it moves some character by almost its full norm.

Lemma 1 (Uniform-essentiality obstruction). *Let $\mathcal{A}$ have a bounded right approximate identity (u_j) . If a bounded family in $\text{LUC}(\mathcal{A})$ has a factorization*

$$S_\lambda = \Psi_\lambda \cdot T, \quad \sup_\lambda \|\Psi_\lambda\| < \infty, \quad T \in \text{LUC}(\mathcal{A}),$$

then

$$\sup_\lambda \|S_\lambda \cdot u_j - S_\lambda\| \longrightarrow 0.$$

Proof. First, $T \cdot u_j \rightarrow T$ in norm. Indeed, by definition $\text{LUC}(\mathcal{A}) = \overline{\text{span}}(\mathcal{A}^* \cdot \mathcal{A})$. For a generator $R \cdot a$,

$$(R \cdot a) \cdot u_j = R \cdot (au_j) \longrightarrow R \cdot a,$$

and boundedness of (u_j) extends this convergence to the closed span. Using the module associativity recorded in the source,

$$(\Psi_\lambda \cdot T) \cdot u_j = \Psi_\lambda \cdot (T \cdot u_j).$$

Consequently, if $C = \sup_\lambda \|\Psi_\lambda\|$, then

$$\sup_\lambda \|S_\lambda \cdot u_j - S_\lambda\| \leq C \|T \cdot u_j - T\| \longrightarrow 0. \quad \square$$

Theorem 1. $L^1(\mathbb{R})$ has an $F\ell^1(\aleph_0)$ -basis, but $\text{LUC}(L^1(\mathbb{R}))$ does not have the $\aleph_0$ -factorization property. Thus the implication asked for in Remark 4.15(iii) is false.

Proof. Let (u_j) be any bounded approximate identity in $L^1(\mathbb{R})$. With the usual dual pairing and convolution product, direct calculation gives

$$\begin{aligned} \langle S_n \cdot u_j, b \rangle &= \langle S_n, u_j * b \rangle \\ &= \left(\int_{\mathbb{R}} u_j(t) e^{int} dt \right) \langle S_n, b \rangle \quad (b \in L^1(\mathbb{R})). \end{aligned}$$

Hence

$$S_n \cdot u_j = m_j(n) S_n, \quad m_j(n) = \int_{\mathbb{R}} u_j(t) e^{int} dt.$$

(The opposite pairing convention merely replaces n by $-n$.) For every fixed j , the Riemann–Lebesgue lemma [2, Ch. 1] yields $m_j(n) \rightarrow 0$ as $n \rightarrow \infty$. It follows that

$$\sup_{n \geq 1} \|S_n \cdot u_j - S_n\|_\infty = \sup_{n \geq 1} |m_j(n) - 1| \geq 1 \quad \text{for every } j.$$

If the requested factorization existed, there would be a bounded sequence $(\Psi_n) \subset \text{LUC}(\mathcal{A})^*$ and a common $T \in \text{LUC}(\mathcal{A})$ with $S_n = \Psi_n \cdot T$. Lemma 1 would make the preceding supremum tend to zero, a contradiction. The source hypothesis was verified above from Theorem 6.1, so this is the desired counterexample. $\square$

Concrete Gaussian check

For the normalized Gaussian approximate identity

$$u_j(t) = \frac{j}{\sqrt{\pi}} e^{-j^2 t^2},$$

one has $m_j(n) = e^{-n^2/(4j^2)}$. Thus already at $n = 20j$ the defect is $1 - e^{-100}$, independently of the scale j . The supplied script checks this numerical illustration; the proof above applies to every bounded approximate identity.

Verification, scope, and novelty

The source hypothesis, target-space identification, module associativity, approximate-identity convergence, Fourier multiplier, and uniform contradiction were audited separately. The result does not affect source Theorem 4.16: one-factorization can hold even though a bounded countable family has no common factor.

A bounded search on 27 August 2026 checked the run indexes, the current official arXiv record, exact phrases from Remark 4.15(iii), the paper title and authors, and combinations of $F\ell^1$, $\text{LUC}(\mathcal{A})$, factorization, characters, and approximate identities. Only the source question was found, and the arXiv record still lists v1. Novelty is plausible but provisional.

References

- [1] M. Filali and J. Galindo, *ℓ^1 -bases, algebraic structure and strong Arens irregularity of Banach algebras in harmonic analysis*, arXiv:2412.15029v1, 2024.
- [2] W. Rudin, *Fourier Analysis on Groups*, Interscience, New York, 1962.